Literature Survey

Project: OptiCrop — Smart Agriculture Crop Recommendation System

1. Introduction

Crop recommendation using Machine Learning has become an active area of research as agricultural data (soil nutrients, climate, and historical yield records) has become more accessible. This survey reviews existing approaches to data-driven crop recommendation systems, the algorithms commonly used, and the gaps this project aims to address.

2. Existing Approaches

Rule-Based / Expert Systems

Early agricultural advisory systems relied on static rule sets defined by agronomists (e.g., if rainfall is within a certain range and soil type is X, recommend crop Y). These systems are easy to interpret but do not adapt to new data and struggle with the complex, non-linear interactions between soil and climate parameters.

Classical Machine Learning Models

Several studies have applied classical supervised learning algorithms - Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Naive Bayes - to tabular agricultural datasets similar to the one used in this project (N, P, K, temperature, humidity, pH, rainfall mapped to a recommended crop). Across this body of work, ensemble tree-based methods such as Random Forest consistently report the highest accuracy (commonly above 99%) due to their ability to capture non-linear, interaction-based decision boundaries between features. Logistic Regression is frequently used as a baseline model because it is fast to train, interpretable, and still achieves strong accuracy (typically 96-98%) on well-separated agricultural datasets.

Deep Learning Approaches

Some recent work has explored Artificial Neural Networks (ANN/MLP) for crop recommendation. However, the consensus across the literature is that deep learning offers little to no advantage over classical ML for small, structured, low-dimensional tabular datasets of this kind (a few thousand rows, seven numeric features). Deep Learning is better suited to high-dimensional, unstructured data such as satellite imagery or remote sensing data, which several studies use for related tasks like crop health monitoring and yield prediction rather than crop selection.

Hybrid and IoT-Integrated Systems

A growing number of systems integrate IoT sensors (for real-time soil moisture, temperature, and nutrient readings) with ML models to provide live, location-specific crop recommendations rather than relying on manually entered or historical data. This represents a natural extension of the approach used in this project.

3. Comparative Summary of Algorithms

Algorithm	Typical Accuracy*	Strengths	Limitations
Logistic Regression	96-98%	Fast, interpretable, good baseline	Assumes linear decision boundaries
Decision Tree	97-99%	Captures non-linear splits, interpretable	Prone to overfitting on small data
Random Forest	99%+	Robust, handles feature interactions well	Less interpretable, slower to train
KNN	96-97%	Simple, no training phase	Slow at prediction time, sensitive to scaling
Naive Bayes	95-99%	Fast, works well with independent features	Assumes feature independence
SVM	97-98%	Effective in high-dimensional space	Needs careful kernel/parameter tuning
ANN / Deep Learning	~96-98%	Can model complex patterns given enough data	Overkill for small tabular datasets; harder to interpret

**Approximate ranges reported across multiple studies on similar crop recommendation datasets; exact figures vary by dataset and preprocessing.*

4. Research Gap

While accuracy on this type of dataset is generally high across most ML algorithms, most existing academic implementations stop at model training and evaluation in a notebook environment. There is limited work on deploying these models as accessible, farmer-facing web tools, particularly for small and marginal farmers who may not have technical expertise. This project (OptiCrop) addresses this gap by combining a trained Logistic Regression model with a simple web interface for real-time crop recommendation.

5. Conclusion

Based on the literature, Logistic Regression is a strong and interpretable baseline model for this problem, while ensemble methods like Random Forest offer marginally higher accuracy at the cost of interpretability. Deep Learning approaches do not show a meaningful advantage for this dataset size and structure. This project adopts Logistic Regression as the core model, with scope for future comparison against Random Forest and other ensemble methods.

6. References

1 Crop Recommendation Dataset - IBM HR Analytics style agricultural dataset (N, P, K, temperature, humidity, pH, rainfall, crop label), widely used in ML coursework and research on precision agriculture.

2 Various peer-reviewed studies and conference papers on ML-based crop recommendation systems (general literature on precision agriculture and crop yield prediction using classical ML algorithms).

3 Scikit-learn documentation - Logistic Regression, Random Forest, Decision Tree, KNN, Naive Bayes, SVM implementations.